

Task 2. Project Report: Application of the Data Vault Method Using the Example of a Receipt

Author: Devendra Shrestha

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1. Introduction

In the era of agile software development, the ability to adapt data warehousing methodologies to rapidly evolving business requirement has become a vital success factor for Business Intelligence (BI) initiatives. Traditional modeling approaches demonstrate effectiveness within structured environments but encounter difficulties adapting to the flexible requirements of contemporary data ecosystems. Data Vault modeling has become a well-regarded option among new methodologies that address current data warehousing challenges because of its strong performance with scalable and flexible systems that require iterative development.

The analysis in this report contrasts Data Vault modeling with the traditional star schema approach through the application of a shopping basket receipt case study. The report aims to analyze the strengths and limitations of both models through a synthesis of existing literature review of Collier (2011), Corr (2011), Hultgren (2012) , and Linstedt and Olschimke (2015), which served as the theoretical foundation for the research and their practical implementation, focusing specifically on their agility, scalability, and implementation complexity.

Data Vault modeling was chosen because its documented benefits enable agile development support and efficient scaling capabilities within complex BI systems. The Data Vault approach enables organizations to reduce restructuring work when business rules evolve which meets the essential needs of modern data warehousing unlike the star schema which provides superior query performance for unchanging dimensional structures. This report investigates how organizations can improve their BI strategies through actionable insights that align with agile principles.

2. Literature Review

2.1 Agile Analytics and Data Warehousing

According to Collier (2011), agile methodologies play a crucial role in BI by promoting iterative development and user collaboration. Both agile principles and Data Vault modeling offer strong support for adaptability to evolving business requirements.

2.2 Dimensional Modeling (Star Schema)

The foundation of data modeling BI traditionally lies in dimensional modeling, particularly the star schema. According to Corr (2011) the star schema is both easy to understand and designed with business users in mind as it features a central fact table and multiple surrounding dimension

tables. This structure supports fast data querying and performs best in static report generation scenarios.

2.3 Data Vault Modeling

Linstedt and Olschimke(2015) argue that data vault is a flexible and scalable method designed for agile environments, organizing the data in hubs (business keys), links (relationship), and satellites (descriptive attributes) to easily adapt to changes without major redesign on data. Hultgren (2012) further emphasizes that Data vault helps to support the historical tracking, making it ideal for enterprise data warehouses that need traceability and auditability.

3. Data Modeling Approaches

3.1 Star Schema Model: Receipt Analysis Example

The Star Schema is a traditional dimensional modeling technique consists of **Central fact table** I surrounded by related **Multiple dimension tables**. For the receipt-based shopping basket analysis, the schema is a follows:

- **Fact Table:**
 - Fact_Receipt_Position(Receipt_ID, Product_ID, Store_ID, Date_ID Quantity, Price)
- **Dimension Tables:**
 - Dim_Product (Product_ID, Product_Name, Category)
 - Dim_Time (Date_ID, Date, Day, Month, Year)
 - Dim_Location(Store_ID, Store_Name, City, Country)

3.1.1 Star Schema Diagram

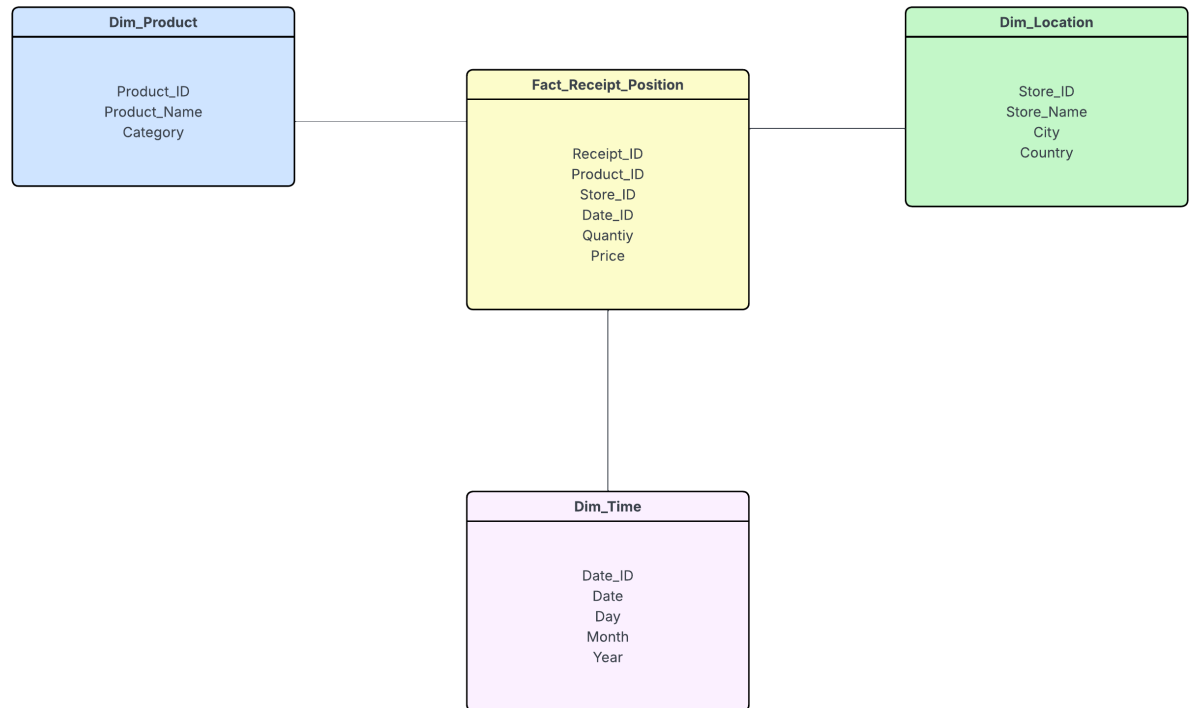


Fig. 1 made by Author's own

3.1.2 Advantages of Using Star Schemas

- It simplifies and speeds up data queries by using a denormalized structure with fewer joins, making aggregation and reporting faster.
- It delivers faster query performance by minimizing joins and optimizing indexing, enabling quick access to insights for decision-makers.
- It makes analysis intuitive by clearly organizing dimensions around facts, simplifying relationships and hierarchies for users.

3.1.3 Disadvantages of Using Star schemas

- It risks data redundancy and integrity issues due to denormalization, leading to potential duplication and stale information.

- It increases storage and compute costs due to redundant data, especially with large datasets.
- It lacks flexibility for complex queries, as they are optimized for specific analytics rather than multi-join scenarios.

3.2 Data Vault Model: Receipt Analysis Example

The data vault is a hybrid data modeling approach that combines the strengths of third normal form and star schema which has vital role in data warehouse that prioritizes, flexibility, scalability, and auditability. The Data Vault model decompose its structure into hubs (business keys), links (relationships), and satellites (descriptive attributes). For the receipt-based shopping basket analysis, the structure is a follows:

Hubs (Business keys):

- **Hub_Product** (ProductID)
- **Hub_Store** (StoreID)
- **Hub_Date** (DateID)

Links (Relationships):

- **Link_Receipt** (ReceiptID, ProductID, StoreID, DateID)

Satellites (Descriptive Data):

- **Sat_ProductDetails** (ProductID, ProductName, Category)
- **Sat_StoreDetails** (StoreID, StoreName, City, Country)
- **Sat_DateDetails** (DateID, Date, Day, Month, Year)

3.2.1 Data Vault Model Diagram

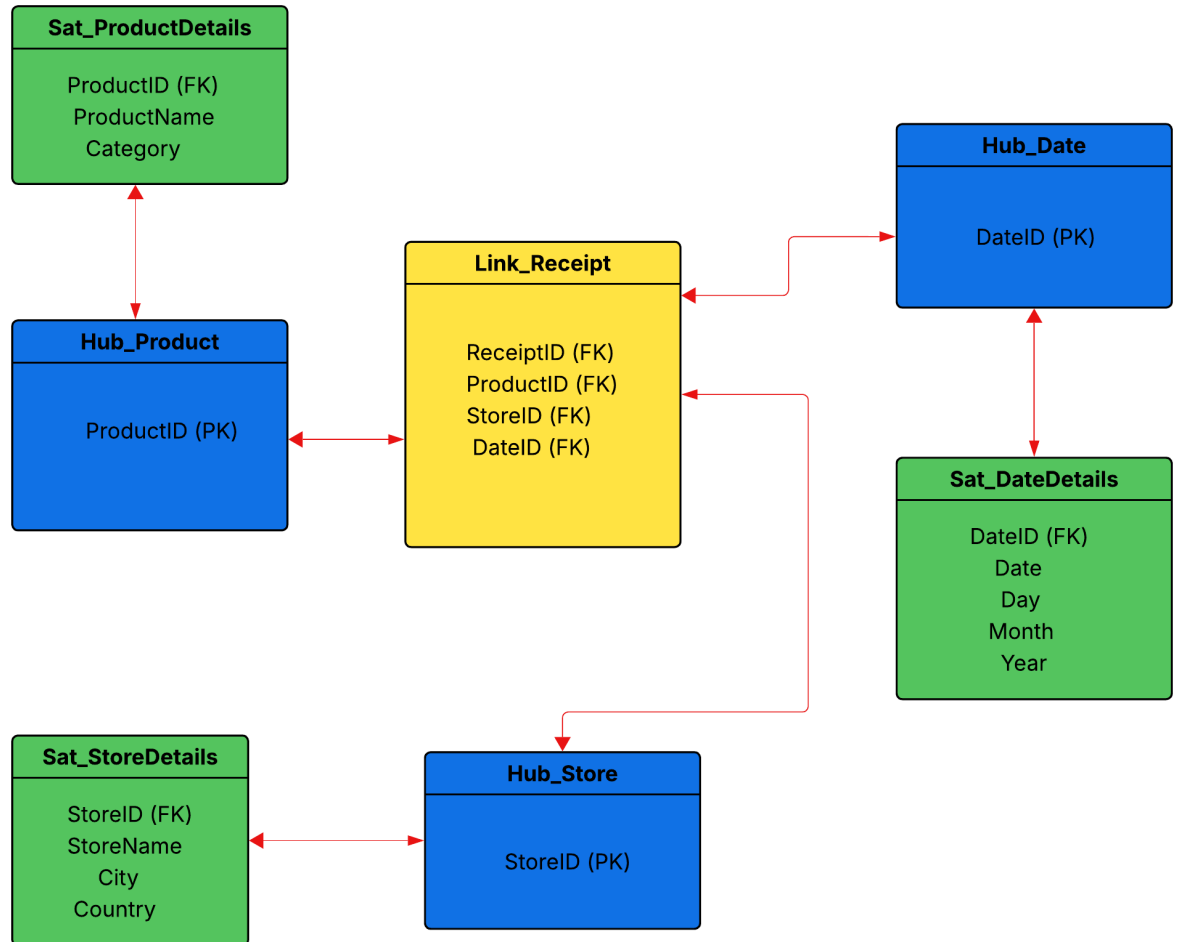


Fig. 2 made by Author's own

3.2.2 Advantages of Using Data Vault Model:

- Data Vault maintains raw source data in its staging area as well as transformed data in hubs, links and satellites which supports traceability and reprocessing without needing to access source systems again.
- It simplifies transformations by using standardized hub-link-satellites structures, reducing complex joins and enabling incremental loading while maintaining agility for evolving business needs.
- Data Vault decouples business rules from raw data by storing them separately in satellites and links, enabling flexible updates without restricting the core mode.

3.2.3 Disadvantages of Using Data Vault Model:

- Data Vault’s historical tracking and non-destructive design inherently require more storage.
- It demands specialized modeling expertise due to its rigid hub-link-satellite architecture, making initial setup more complex than traditional warehouse despite long-term flexibility benefits.
- Data Vault shifts transformation complexity to downstream data marts, requiring intricate joins across hubs, links, and satellites, which can impact query performance for end-users.

4. Comparative Analysis

Aspect	Star Schema	Data Vault
Flexibility	Low (rigid structure)	High (adaptable)
Query Performance	Fast (denormalized)	Slower (normalized)
Historical Tracking	Limited	Strong(built-in history)
Ease of Use	Simple for analysts	Complex (requires expertise)
Scalability	Challenging for growth	Highly scalable

5. Conclusion

This report compared Star Schema and Data Vault modeling approaches while analyzing receipt-based shopping basket data. The comparative assessment reveals that both data warehousing models offer significant benefits yet their applicability relies on specific business context and project requirements.

The Star Schema provides optimal performance for quick query execution and straightforward operation. The denormalized structure of this model enables fast data access and streamlined analysis which makes it well-suited for static reporting in setups with unchanging business rules. The inflexible structure of this model restricts its adaptability and leads to data redundancy issues along with scalability difficulties when data complexity increases.

Data Vault modeling stands out with its exceptional flexibility alongside scalability and agile development support capabilities. The normalized and modular structure of Data Vault uses hubs, links, and satellites to enable historical data capture and adaptability to business changes without impacting the core data model. The Data Vault approach adds complexity to design and downstream transformations but establishes a solid base for BI systems that need long-term agility and auditability.

In conclusion, Organizations developing data warehouse solutions that need scalability and future readiness while maintaining agile alignment should evaluate Data Vault modeling as a serious option. The Star Schema continues to serve as an appropriate solution for analytical projects that demand high performance and simplicity within stable environments. Organizations should base their choice between these models on strategic priorities including adaptability requirements, performance needs, historical tracking capability, and technical expertise.

6. References

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